Assignment: Data Modelling & Relationship Verification using Matrix Visual

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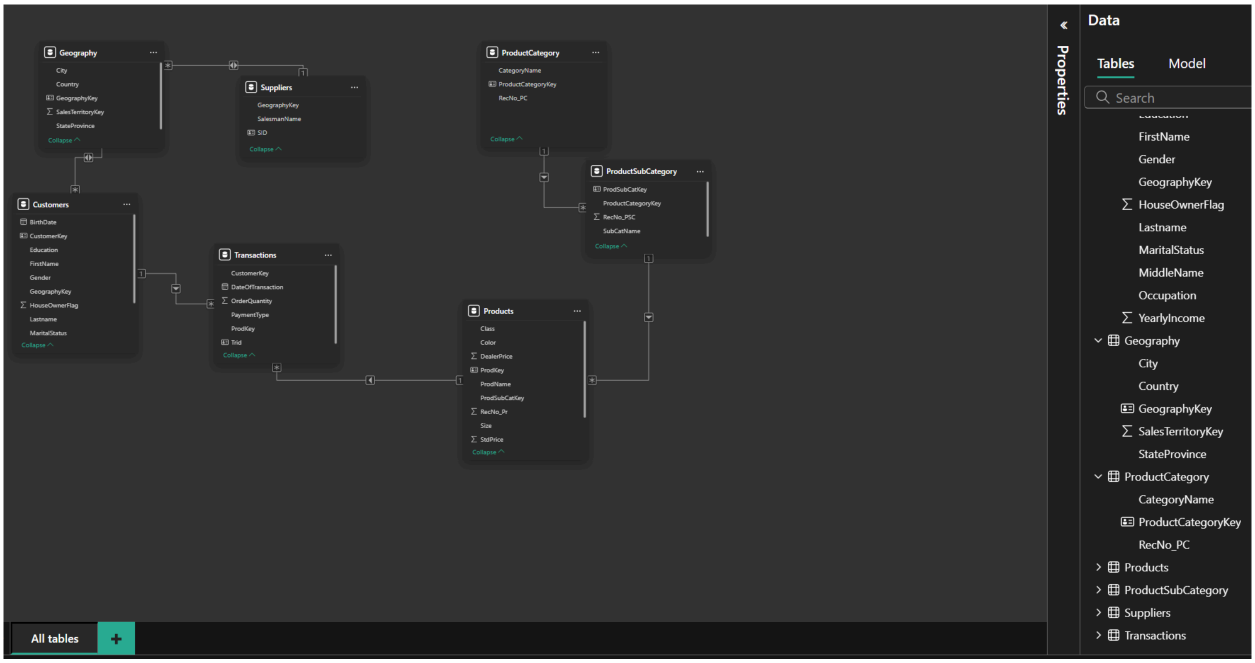
Part 1: Model Creation

- Screenshot of the Data Model view showing all relationships.




| From: table (column) | Relationship | To: table (column) | Status |
|---|---|---|---|
| Customers (GeographyKey) | * — 1 | Geography (GeographyKey) | Active |
| Geography (GeographyKey) | * — 1 | Suppliers (GeographyKey) | Active |
| Products (ProdSubCatKey) | * — 1 | ProductSubCategory (ProdSub… | Active |
| ProductSubCategory (Product… | * — 1 | ProductCategory (ProductCate… | Active |
| Transactions (CustomerKey) | * — 1 | Customers (CustomerKey) | Active |
| Transactions (ProdKey) | * — 1 | Products (ProdKey) | Active |

Part 3: Relationship Verification

**Screenshot of both matrix visuals with drill-down enabled.**

1. Screenshots 1 and 2 and 3 Matrix 1 with drill down enabled

| Year | 2010 | | **Total** | |
|---|---|---|---|---|
| CategoryName | Sum of OrderQuantity | Sum of TotalPrice | **Sum of OrderQuantity** | **Sum of TotalPrice** |
| ⊞ Clothing | 29545 | 42,872.50 | **29545** | **42,872.50** |
| ⊞ Accessories | 16821 | 25,395.00 | **16821** | **25,395.00** |
| ⊞ Electronics | 14232 | 10,26,550.00 | **14232** | **10,26,550.00** |
| **Total** | **60598** | **10,94,817.50** | **60598** | **10,94,817.50** |

| Year | 2010 | | **Total** | |
|---|---|---|---|---|
| ProdSubCatKey | Sum of OrderQuantity | Sum of TotalPrice | **Sum of OrderQuantity** | **Sum of TotalPrice** |
| PSCK4 | 14090 | 16,015.00 | **14090** | **16,015.00** |
| PSCK1 | 9181 | 5,22,675.00 | **9181** | **5,22,675.00** |
| PSCK6 | 9026 | 13,065.00 | **9026** | **13,065.00** |
| PSCK5 | 6429 | 13,792.50 | **6429** | **13,792.50** |
| PSCK8 | 6182 | 18.615.00 | **6182** | **18.615.00** |
| **Total** | **60598** | **10,94,817.50** | **60598** | **10,94,817.50** |

| Year | 2010 | | **Total** | |
|---|---|---|---|---|
| CategoryName | Sum of OrderQuantity | Sum of TotalPrice | **Sum of OrderQuantity** | **Sum of TotalPrice** |
| ⊟ **Electronics** | **5027** | **2,43,875.00** | **5027** | **2,43,875.00** |
| ⊟ **PSCK2** | **5027** | **2,43,875.00** | **5027** | **2,43,875.00** |
| Samsung | 3102 | 1,11,750.00 | **3102** | **1,11,750.00** |
| Nokia | 1925 | 1,32,125.00 | **1925** | **1,32,125.00** |
| **Total** | **5027** | **2,43,875.00** | **5027** | **2,43,875.00** |

2. Screenshots 4 and 5 Matrix 2 with drill down enabled

| Year | 2010 | | Total | |
|---|---|---|---|---|
| Country | Total Sales | Count of CustomerKey | **Total Sales** | **Count of CustomerKey** |
| ⊞ | 43,15,14,797.50 | 17484 | **43,15,14,797.50** | **17484** |
| ⊞ United States | 67,18,582.50 | 236 | **67,18,582.50** | **236** |
| ⊞ United Kingdom | 56,47,257.50 | 197 | **56,47,257.50** | **197** |
| ⊞ France | 56,40,935.00 | 200 | **56,40,935.00** | **200** |
| ⊞ Germany | 54,64,547.50 | 200 | **54,64,547.50** | **200** |
| ⊞ Australia | 33,17,812.50 | 111 | **33,17,812.50** | **111** |
| ⊞ Canada | 16,05,530.00 | 56 | **16,05,530.00** | **56** |
| **Total** | **45,99,09,462.50** | **18484** | **45,99,09,462.50** | **18484** |

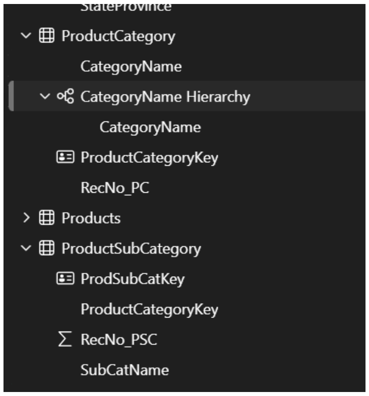
| Year | 2010 | | Total | |
|---|---|---|---|---|
| Country | Total Sales | Count of CustomerKey | **Total Sales** | **Count of CustomerKey** |
| ⊟ **Germany** | **26,02,400.00** | **98** | **26,02,400.00** | **98** |
| ⊟ **Bayern** | **2,43,750.00** | **9** | **2,43,750.00** | **9** |
| Erlangen | 27,200.00 | 1 | **27,200.00** | **1** |
| Frankfurt | 53,350.00 | 2 | **53,350.00** | **2** |
| Grevenbroich | 27,200.00 | 1 | **27,200.00** | **1** |
| Hof | 81,600.00 | 3 | **81,600.00** | **3** |
| Ingolstadt | 54,400.00 | 2 | **54,400.00** | **2** |
| ⊞ **Brandenburg** | **54,400.00** | **2** | **54,400.00** | **2** |
| ⊞ **Hamburg** | **5,30,150.00** | **20** | **5,30,150.00** | **20** |
| ⊞ **Hessen** | **5,71,200.00** | **21** | **5,71,200.00** | **21** |
| ⊞ **Nordrhein-Westfalen** | **5,77,400.00** | **22** | **5,77,400.00** | **22** |
| ⊞ **Saarland** | **6,25,500.00** | **24** | **6,25,500.00** | **24** |
| **Total** | **26,02,400.00** | **98** | **26,02,400.00** | **98** |

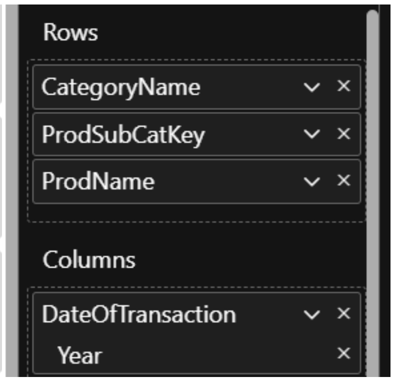
**1. Why certain relationships needed bidirectional filtering.**

Bidirectional filtering was applied between Customers-Geography and Suppliers-Geography because these relationships require context flow in both directions. When analyzing customer data, we need geographical context to flow back to customer filters, and when examining geographical data, customer information should influence the results. This two-way filtering ensures that selecting a specific geography filters customers appropriately, and selecting customers filters geography correctly. Without bidirectional filtering, certain analytical scenarios would produce incomplete or incorrect results.

**2. How hierarchies help in data analysis.**

Hierarchies provide structured drill-down capabilities that enable users to analyze data at different levels of granularity. **However, Power BI hierarchies have a critical limitation: they can only be created within a single table.** In this dataset, while the Geography Hierarchy works perfectly (all fields in one table), the logical Product Hierarchy (Category → SubCategory → Product) cannot be implemented as a formal hierarchy due to the normalized snowflake design. Instead, analysts must use multiple separate fields in matrix visuals to achieve similar drill-down functionality, which requires manual arrangement of fields in the correct order. This limitation demonstrates why some organizations choose to denormalize their product dimensions into a single table specifically for Power BI implementations, trading storage efficiency for improved user experience and analytical capability.





3. **What considerations are made to apply some relationships and filters.**

Relationship design must consider cardinality requirements, with most relationships being one-to-many to ensure proper aggregation and avoid data duplication. Filter direction choices depend on analytical requirements - single direction for performance optimization, bidirectional where cross-filtering is essential for business logic. Data integrity considerations include ensuring foreign keys exist in related tables and managing inactive relationships to prevent ambiguous filter paths. Performance implications guide decisions about relationship activation and filter complexity.

4. **Write a note on snowflake and star schema in terms of the given dataset**

This dataset implements a hybrid star/snowflake schema design where Transactions serves as the central fact table connected to multiple dimension tables. The pure star schema elements include direct connections from Transactions to Customers and Geography. However, the snowflake elements appear in the Product dimension chain (ProductCategory → ProductSubCategory → Products), creating a normalized structure that reduces redundancy but adds complexity. While this snowflake design is efficient for storage and maintains data integrity, it creates a significant limitation in Power BI: **you cannot create a single hierarchy spanning multiple tables**. This means the logical product hierarchy (Category → SubCategory → Product) must be implemented through multiple separate fields in matrix visuals rather than a formal Power BI hierarchy, requiring more complex navigation and potentially impacting user experience. This demonstrates a key trade-off in snowflake schemas - while they excel in normalized data storage, they can limit some analytical tool capabilities that work better with denormalized star schema designs.

5. **Justify the relationship cardinality you made between tables (many-to-one and one-to-many).**

The one-to-many relationships were established based on natural business hierarchies and transactional logic. ProductCategory to ProductSubCategory (one category contains many subcategories), Products to Transactions (one product appears in many sales transactions), and Customers to Transactions (one customer makes many purchases) represent the core business relationships. These cardinalities ensure proper aggregation behavior in visuals, prevent data duplication, and maintain referential integrity. The many-to-one relationships like Customers to Geography reflect that multiple customers can reside in the same geographical location.

6. **Justify why a bidirectional filter is applied for specific relationship**

Bidirectional filtering was specifically applied to Customer-Geography and Supplier-Geography relationships because these connections require mutual context sharing for comprehensive analysis. When filtering by geographical regions, users need to see which customers and suppliers are affected, and conversely, when analyzing specific customers or suppliers, their geographical context becomes crucial. This bidirectional flow enables complex analytical scenarios like "show sales performance in regions where our top customers are located" or "analyze supplier performance by geographical proximity to customers." Without this bidirectional capability, such cross-dimensional analysis would be limited or impossible.